

Candidate Name _____

Civics Group Reg Number

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MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2011
Higher 1

Answers

H1 BIOLOGY

Paper 1 (Multiple Choice Questions)

8875/1

22 September 2011

1 Hour

Additional Materials: OMR

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, civics group and register number in the spaces at the top of this page and on the OMR sheet

There are **thirty** questions in this section. Answer **all** questions. For each question, there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate **OMR** answer sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

Calculators may be used.

At the end of the examination,

1. Write your name, index number and civics group on the OMR sheet and shade accordingly.
2. Hand the OMR only.

This paper consists of 18 printed pages

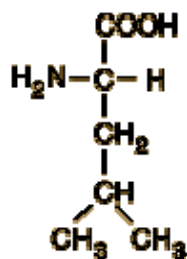
MJC Prelim 2011

H1 Bio P1

Answer Scheme

1. C	7. B	13. D	19. C	25. A
2. C	8. A	14. B	20. D	26. A
3. C	9. B	15. D	21. C	27. D
4. B	10. A	16. A	22. C	28. C
5. C	11. D	17. A	23. B	29. A
6. C	12. A	18. D	24. C	30. B

- 1 The diagram shows a molecule which is an important component of living organisms.

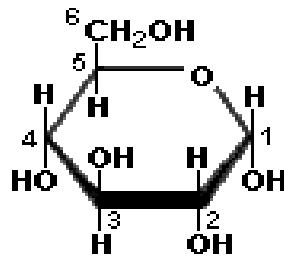


A forensic sample is tested for the presence of a polymer of this molecule.

Which polymer is tested for?

- A DNA
- B Triglyceride
- C Protein**
- D Starch

- 2 A glucose molecule in the diagram has its six carbon atoms numbered.



Which of the following statement is false?

- A Carbon 1 and 4 are involved in the formation of amylose
 - B Carbon 1 and 6 are involved in the formation of amylopectin
 - C Carbon 1 and 4 are involved in the formation of cellulose**
 - D Carbon 1 and 6 are involved in the formation of starch
- 3 Which of the following is/are the most likely consequence for a cell lacking functional Golgi bodies?
- 1. The cell dies because it is unable to make glycoproteins to detect stimuli from its environment.
 - 2. The cell dies from a lack of enzymes to digest food taken in by endocytosis.
 - 3. The cell dies from the accumulation of worn-out organelles within itself.
 - 4. The cell is unable to reproduce itself
 - 5. The cell is unable to export its enzymes or peptide hormones.
- A 1 and 5
 - B 2 and 3
 - C 3, 4 and 5**
 - D 1,2, 3 and 5
- 4 The components of the membranes found within and at the surface of cells contribute to their functions. Examples of components are shown below.
- 1 cholesterol

- 2 glycolipids
- 3 glycoproteins
- 4 phospholipids
- 5 proteins

Which row matches the components with their functions?

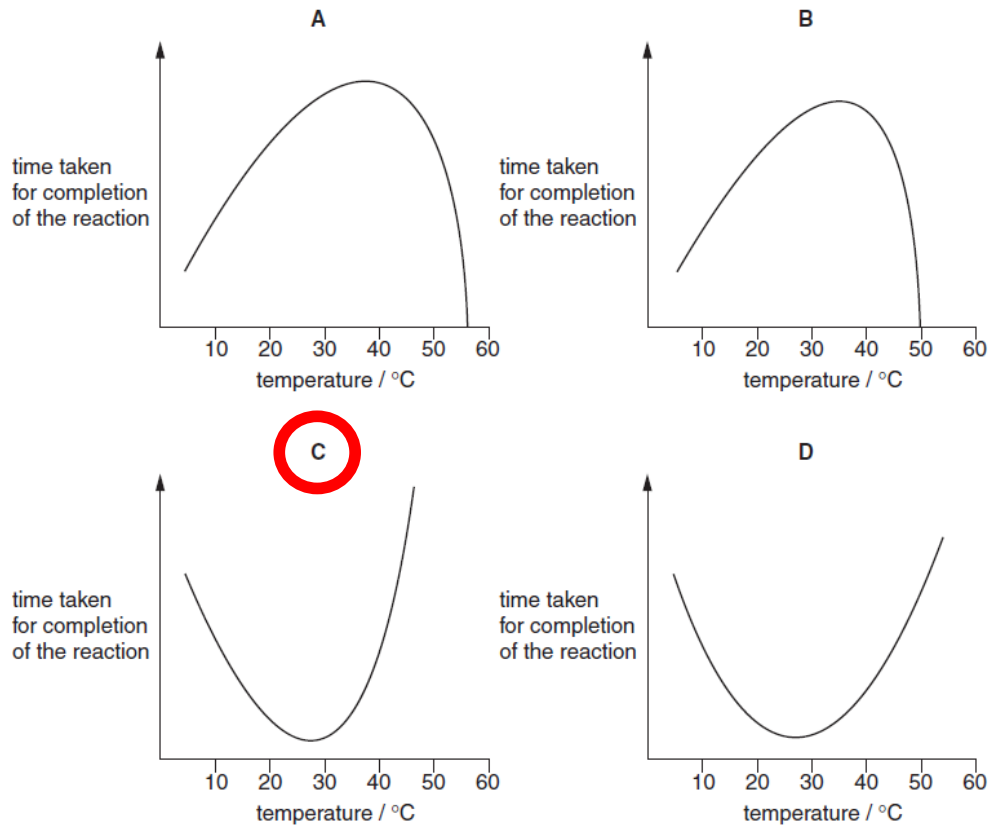
	Binding sites for toxins	Cell-to-cell recognition	Separating dissolved ions	Stabilising the hydrophobic layer	Transporting ions through membranes
A	1	3	5	2	4
B	2	3	4	1	5
C	3	1	2	5	4
D	3	2	1	4	5

5 Which combination would decrease the fluidity of a cell surface membrane?

	saturated fatty acids	non-saturated fatty acids	cholesterol
A	decrease	increase	decrease
B	decrease	increase	increase
C	increase	decrease	decrease
D	increase	decrease	increase

- 6 An enzyme is completely denatured at 50°C. A fixed concentration of this enzyme is added to a fixed concentration of its substrate. The time taken for completion of the reaction is measured at different temperatures.

Which graph shows the results?



- 7 A student obtained a sample of DNA. mRNA was transcribed from this DNA and the two samples were subsequently purified. He then separated the two strands of the DNA sample. The base compositions of each strand and that of the mRNA were analysed. The results of the analysis are shown in the table below.

	A	G	C	T	U
DNA strand 1	19.1	26.0	31.0	23.9	0.0
DNA strand 2	24.2	30.8	25.7	19.3	0.0
DNA strand 3	20.5	25.2	29.8	24.5	0.0
mRNA	19.0	25.9	30.8	0.0	24.3

Which strand of the DNA serves as a template for mRNA synthesis?

- A Strand 1
 - B Strand 2**
 - C Strand 3
 - D Strand 2 and 3
- 8 What is the basis for the difference in the synthesis of the leading and lagging strands of DNA molecules?

- I The anti-parallel arrangement of the DNA strands.
- II The RNA primers are required to initiate DNA elongation.
- III DNA polymerase can join new nucleotides to the 3' end of the growing strand.
- IV Helicase and single-stranded binding proteins work at the 5' end of the DNA strand.

A I and III

B I and IV

C II and III

D II and IV

- 9 An experiment is carried out to show that DNA replication is semi-conservative. *E.coli* cells whose DNA contains only 'heavy' nitrogen (^{15}N) are grown in a culture medium containing the normal isotope of nitrogen (^{14}N) and allowed to divide three times.

What will be the percentage of DNA molecules containing both forms of nitrogen after the third division?

- A 12.5% **B 25%** C 50% D 100%

- 10 Which correctly describes the structure of DNA during transcription?

A Double-stranded with single-stranded sections associated with polymerase enzymes.

B Double-stranded region with two loops of unpaired bases.

C Double strand held together by covalent bonds between nucleotide pairs.

D Single strand formed by base pairing cytosine with guanine and adenine with thymine.

- 11 Which of the following observations proves that the anticodon of a tRNA molecule, and not the amino acid that it carries, recognizes and binds to the mRNA codon at the ribosome?

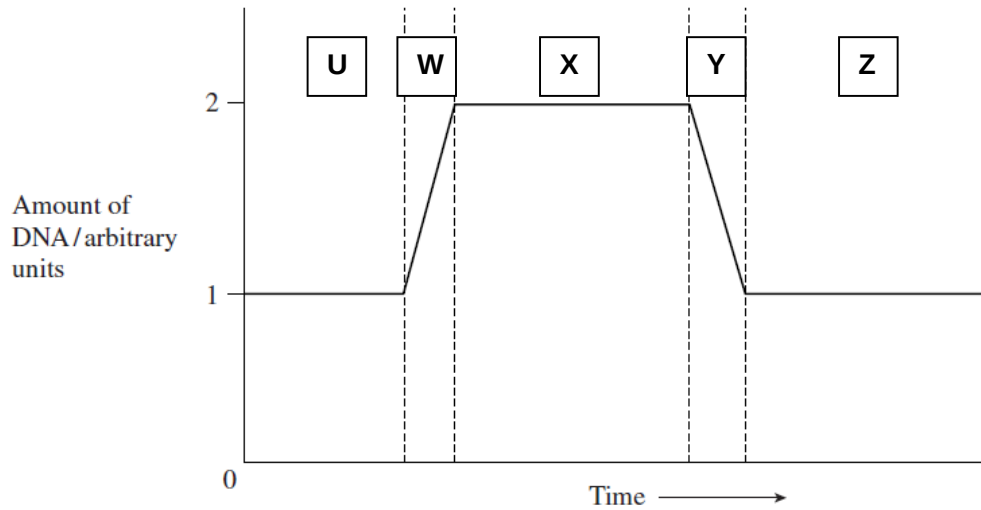
A A tRNA carrying a valine but with an isoleucine anticodon does not place any amino acid onto a growing peptide chain when an isoleucine mRNA codon is present.

B A tRNA carrying a valine with a valine anticodon places a valine onto a growing peptide chain when only the first two bases of the anticodon pairs with the mRNA codon.

C A tRNA carrying a valine with a valine anticodon does not place any amino acid onto a growing peptide chain when only the first two bases of the anticodon pairs with the mRNA codon.

D A tRNA carrying a valine but with an isoleucine anticodon places a valine onto a growing peptide chain when an isoleucine mRNA codon is present.

- 12 The graph shows changes in the amount of DNA in a cell during one cell cycle. The letters U – Z marks out the different phases in the cell cycle.



Many drugs that are used to treat cancer work at different time periods during the cell cycle.

- (i) Cisplatin binds to DNA and stops free DNA nucleotides from joining together.
- (ii) Drug B stops spindle fibres from shortening.

With reference to the cell cycle above, determine when these 2 drugs work.

	Cisplatin	Drug B
A	W	X
B	W	Y
C	U	X
D	U	Z

- 13 In rats, the allele of a gene for 'mottled' coat (M) and the recessive allele (m) for 'normal' coat are sex-linked. The allele of a gene for 'long' whiskers (W) and the recessive allele (w) for 'short' whiskers are autosomal.

A male rat with a normal coat and short whiskers was mated on several occasions to the same female. The offspring showed the following phenotypes in equal proportions.

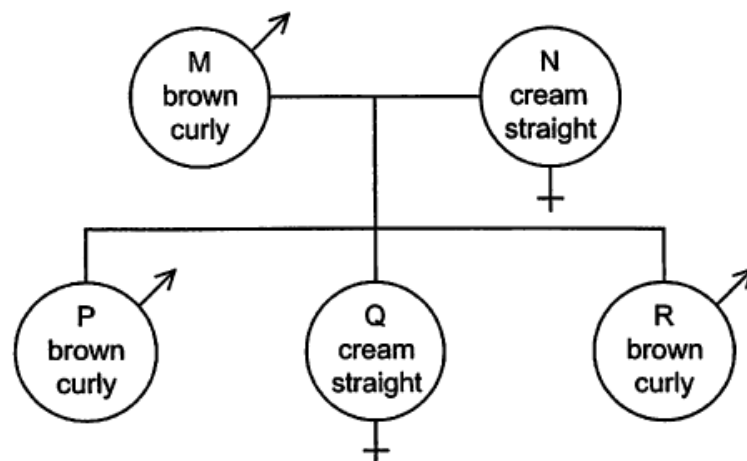
- Mottled females and males with long whiskers
- Mottled females and males with short whiskers
- Normal females and males with long whiskers
- Normal females and males with short whiskers

What are the genotypes of the parents?

- A X^mYww and X^MX^MWW
- B X^mY^mww and X^MX^MWw
- C X^mY^mww and X^MX^mWW
- D X^mYww and X^MX^mWw**

- 14 Assume that in goats, a pair of alleles is responsible for the inheritance of hair color and that another pair controls hair texture. These pairs are located on different autosomes. The allele for brown hair (B) is dominant to the allele for cream hair (b), and the allele for curly hair (C) is dominant to the allele for straight hair (c).

The diagram shows a cross between two goats.



If R is mated with a female goat of the same genotype as M, what is the probability that the first offspring being a male with cream colored, straight hair?

- A 0
B 1 in 32
 C 1 in 16
 D 1 in 4

15 Two genes involved in coat color of goats are at loci on different chromosomes.

The color gene C causes the hairs to be uniform color and has three alleles.

- C^{DB} giving dark brown hairs
- C^B giving black hairs
- C^{MB} giving medium brown hairs

A dominant allele of the agouti gene (A^G) causes the development of white hairs between the colored hairs giving the coat a shaded appearance.

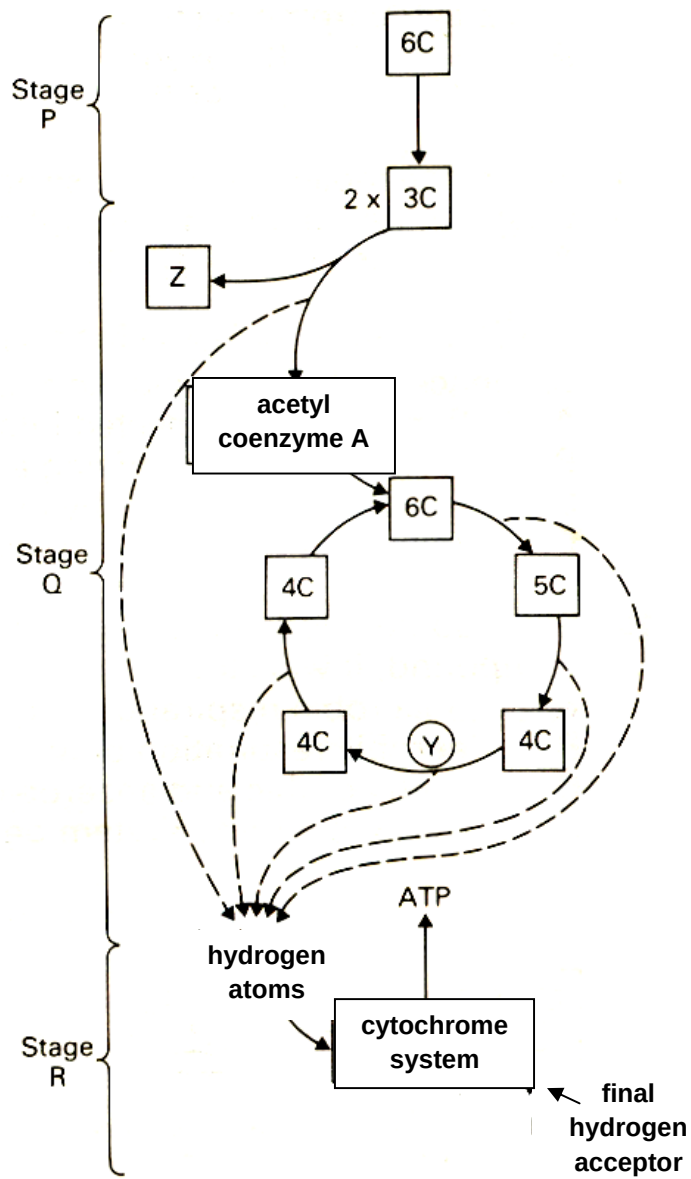
The table shows the results of crosses between a male goat and two female goats.

Cross	Parents	Offspring
1	Black agouti male x Uniformly dark brown female	50% uniform, 50% agouti; 50% dark brown, 50% black
2	Black agouti male x Medium brown agouti female	all agouti; 50% black, 50% medium brown

Which row shows the possible genotypes of the male and female goats?

	Black agouti male	Uniformly dark brown female	Medium brown agouti female
A	$C^B C^{MB} A^G A^G$	$C^B C^{DB} A^G A^g$	$C^{MB} C^{MB} A^G A^G$
B	$C^B C^{MB} A^G A^g$	$C^B C^{DB} A^g A^g$	$C^B C^{MB} A^G A^G$
C	$C^B C^{DB} A^G A^G$	$C^{DB} C^{MB} A^G A^g$	$C^B C^{DB} A^G A^g$
D	$C^B C^{MB} A^G A^g$	$C^B C^{DB} A^g A^g$	$C^{MB} C^{MB} A^G A^G$

16 The diagram shows a biochemical pathway in a typical eukaryotic cell.



Which of the following indicates the correct locations at which stages P, Q and R occur?

	Cristae	Cytosol	Mitochondrial matrix
A	R	P	Q
B	Q	P	R
C	P	R	Q
D	R	Q	P

17 Which statement about anaerobic respiration is correct?

- A** From one molecule of glucose, ethanol and lactate production yield the same amounts of ATP
- B** From one molecule of glucose, ethanol production yields less energy than lactate production.
- C** From one molecule of glucose, ethanol production yields more energy than lactate production.
- D** Yeast is able to respire ethanol for the production of ATP.

18 The Calvin cycle can be divided into three phases, involving several reactions.

1. carbon dioxide uptake
2. carbon reduction
3. regeneration

Which of the following correctly matched the reactions with three phases?

	ATP hydrolysis	Formation of NADP	Formation of 3C sugar	Use of RuBP
A	1	3	2	1
B	1, 3	2	1, 2	3
C	2	1, 2	2, 3	3
D	2, 3	2	2	1

- 19** In light, Mg^{2+} ions move into the stroma. In the dark, the concentration of Mg^{2+} ions in the stroma is low.

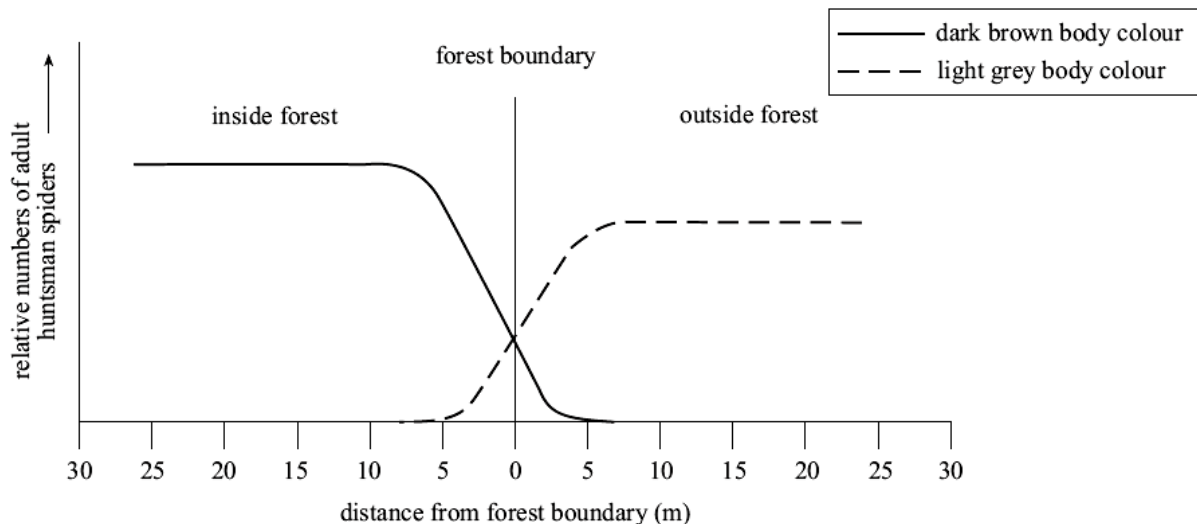
In high Mg^{2+} concentrations, CO_2 and Mg^{2+} react with the active site of RuBP carboxylase (rubisco), making a carbamate group (CO_2NH) and making the rubisco active. In low Mg^{2+} concentrations the carbamate group dissociates making rubisco inactive.

Inactive rubisco will bind tightly at its active site with any RuBP present, so that no rubisco catalysis can take place. An ATPase enzyme called rubisco activase, can, in high concentrations of Mg^{2+} , release the RuBP from the rubisco, producing a carbamate group and activating the rubisco.

Which statement is supported by these facts?

- A** Inactivated rubisco can be reactivated by NADPH made in the light-dependent reaction.
- B** Low concentrations of 3C compound in the stroma of the chloroplast deactivate rubisco.
- C** Rubisco without the carbamate group is inhibited by low concentrations of RuBP.
- D** The rate of carbon dioxide fixation increases when the carbamate group dissociates from rubisco.

20 The graph below shows the distribution of huntsman spiders at a forest boundary.



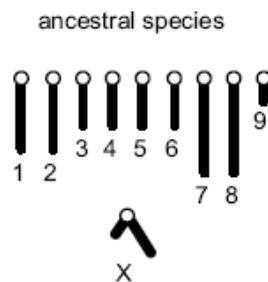
One species of huntsman spider (*Isopeda isopedella*) varies in body colour from dark brown to light grey. In one community at the forest boundary, two populations of this species were found. Some were found living in the leaf litter inside the forest and others were found living in the grass just outside the forest. The relative numbers of dark brown adult spiders and light grey adult spiders found at certain distances from the forest boundary are shown in the graph above.

The best explanation for this distribution is that

- A the two populations of spiders were once different species.
- B the two populations of spiders were unable to interbreed and individuals were adapting to suit their habitats.
- C the differences in the two habitats had changed the physical appearance of individual spiders.
- D a particular body colour provided a selective advantage to spiders in a particular habitat.

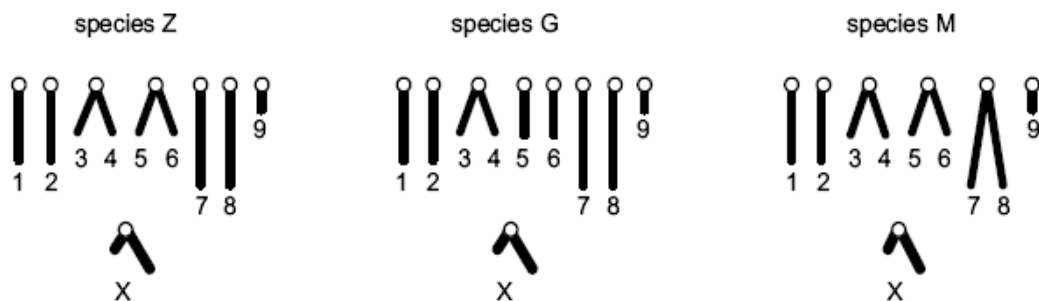
- 21** In some Australian insects, new species have arisen through changes that occurred to chromosomes in an ancestral species. Such changes may involve the joining together of chromosomes, the loss of whole or parts of chromosomes, and rearrangement of the genetic material within chromosomes.

One ancestral species has the following haploid set of chromosomes.



As the changes in chromosomes accumulate, a number of different species can result from a single ancestral species.

Three species that have evolved from the ancestral species shown above have the haploid sets of chromosomes shown below.



The most likely order of evolution of these species is

- A ancestral species, species Z, species G, species M.
- B ancestral species, species Z, species M, species G.
- C ancestral species, species G, species Z, species M.**
- D ancestral species, species G, species M, species Z.

- 22** A population of zooplankton is exposed to a small number of predatory fish that feed on the larger-sized adult zooplankton. Which of the following predictions would most likely occur based on the principles of natural selection?
- A** The predatory fish will evolve smaller mouths so that they do not drive their prey to extinction.
 - B** The small population of predatory fish will grow in number because of abundant food supply in the zooplankton.
 - C** Adult zooplankton will start to reach sexual maturity when they are still relatively small.
 - D** More mutations will be observed in the zooplankton population to ensure a greater chance of survival.

- 23** The incidence of infection with *Staphylococcus aureus* resistant to the antibiotic methicillin is rising yearly.

What is a possible evolutionary reason for this?

- A** natural selection for reduced immunity in the human population
- B** natural selection for resistant *Staphylococcus* in response to continued methicillin use
- C** natural selection of resistant *Staphylococcus* in response to reduced methicillin use
- D** natural selection of *Staphylococcus* resistant to other antibiotics

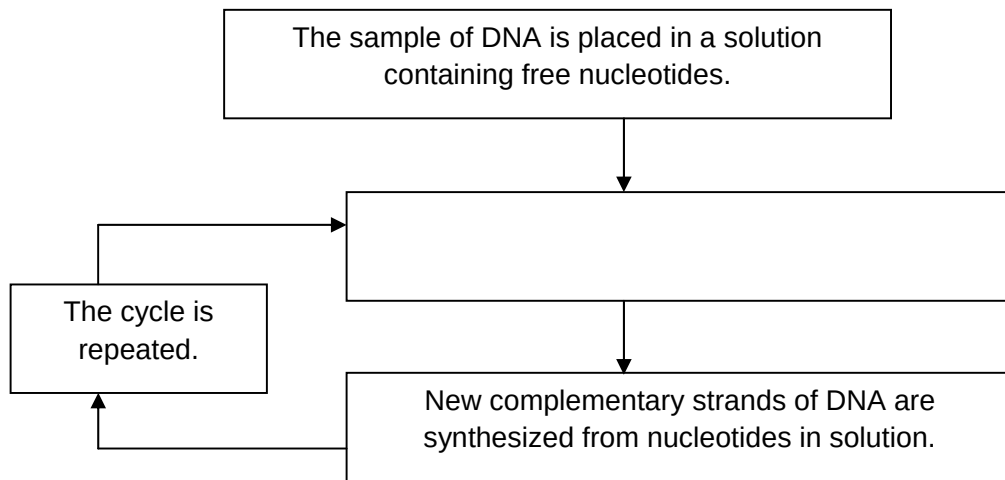
- 24** Corn is a major crop grown in Europe. In the past, it was either ruined by attack from the corn borer larvae or intensively sprayed with pesticides each year. The biotech company Novartis gained approval to insert a *Bt* gene into corn. This gene codes for a protein that kills the larvae feeding on the corn. The genetically engineered *Bt* corn initially thrived without the addition of any pesticides, but later suffered damage from pests again.

Which of the following are possible reasons why the *Bt* corn is again under attack from pests?

I	A strain of resistant larvae has emerged as a chance mutation. The frequency of the gene conferring resistance as well as the number of larvae with resistance has decreased.
II	Corn is being attacked by other pests which are not killed by the protein encoded by <i>Bt</i> gene.
III	Mutations in corn may have led to the loss of the <i>Bt</i> allele. This loss of <i>Bt</i> allele has had a selective advantage in seed fertility.

- A** I and II only
B I and III only
C II and III only
D III only

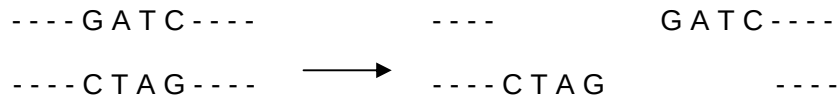
25 The polymerase chain reaction is summarized in the flowchart below.



Which statement completes the flow chart?

- A** Complementary strands of DNA are separated.
- B** Free nucleotides join on the end of DNA strands.
- C** Small sections of DNA are formed.
- D** Strands of DNA bind to RNA primers.

26 The restriction enzyme, *Sau* 3A, cuts DNA as shown.



Which of the plasmids below, with specific regions indicated, can be cut using restriction enzyme *Sau* 3A in order to allow DNA to be inserted into the circular plasmid without loss of any part of the plasmid?

A	---- T A G A T C A C ---- C T C A A G C T ---- ---- A T C T A G T G ---- G A G T T C G A ----
B	---- T T A G A T C A ---- C G A T C G G T ---- ---- A T T C T A G T ---- G C T A G C C A ----
C	---- G A T C G A T C ---- T C C A C G T A ---- ---- C T A G C T A G ---- A G G T G C A T ----
D	---- A A T C A G G T ---- C C A C C T G A ---- ---- T T A G T C C A ---- G G T G G A C T ----

27 Which of the following processes could improve quality of the crop?

- W inserting genes for insect pest resistance into potato
- X inserting genes for vitamin A production into rice
- Y inserting genes for nitrogen fixation into soya beans
- Z inserting genes for drought resistance into rice

A W and X **B** X and Y **C** Y and Z **D** Z and W

28 Scientists are concerned about the escape of genetically modified salmon into the wild.

What is the most likely reason for this concern?

- A** The genetically modified salmon may not survive in the wild.
- B** The mutation rate of the genetically modified salmon will increase.
- C** The genetically modified salmon may replace the wild salmon population.
- D** The growth rate of the genetically modified salmon will be affected.

29 Some of the goals of the Human Genome Project are:

- To determine the sequence of the entire human genome
- To identify all the genes in the human genome
- To find the locus of all the genes on the 46 human chromosomes

Which of the following are ethical concerns arising from the goals stated?

1. Anthropologist tracing the ancestry of human populations.
2. Parents choosing embryos for implantation only after tests for acceptable genes.
3. Insurance company offering cheaper rates to people with genetic disposition to fewer diseases.
4. Scientists developing tests for only some disease-causing genes.
5. Genetic counselors giving advice to people who are genetically pre-disposed to risks.

- A** 2 and 3 **B** 3 and 4 **C** 1 and 5 **D** 4 and 5

30 It has been found that stem cells transformed from intestinal lining to the bone marrow produce all of the different types of blood cell instead of intestinal cell.

Which statement explains this?

- A** All stem cells are totipotent.
- B** Environmental factors change the expression of specific genes.
- C** Specific genes are destroyed by endonucleases.
- D** Specific genes are hidden by condensation of some chromosomes.

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